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Split via structure for semiconductor device packaging

Abstract

A semiconductor device assembly including a substrate; a first split via including a first via land that is disposed on a surface of the substrate and that has a first footprint with a half-moon shape with a first radius of curvature, and a first via that passes through the substrate and that has a second radius of curvature, wherein the first via is disposed within the first footprint; and a second split via including a second via land that is disposed on the surface of the substrate and that has a second footprint with the half-moon shape with the first radius of curvature, and a second via that passes through the substrate and that has the second radius of curvature, wherein the second via is disposed within the second footprint, wherein the first and second via lands are disposed entirely within a circular region having the first radius of curvature.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure generally relates to semiconductor device packaging, and more particularly relates to a split via structure configured for packaging semiconductor devices.

BACKGROUND

(2) Microelectronic devices, such as memory devices, microprocessors, and other electronics, typically include one or more semiconductor dies mounted to a substrate and encased in a protective covering. The semiconductor dies include functional features, such as memory cells, processor circuits, interconnecting circuitry, etc. Semiconductor die manufacturers are under continuous pressure to reduce the volume occupied by semiconductor devices while increasing the capacity and/or speed of the resulting semiconductor assemblies. To meet these demands, semiconductor die manufacturers often stack multiple semiconductor dies vertically on top of each

other and increasingly tightly pack bonding sites and rerouting structures to increase the capacity and/or the performance of semiconductor devices within a limited area on a circuit board or other element to which the semiconductor devices and/or assemblies are mounted. As components are stacked and processing demands increase, the semiconductor devices often include one or more metallization layers with varying trace paths that interconnect through vias. The varying trace paths can help equalize an effective signal travel path between signal sources and destinations to help maintain necessary parity in the timing of signals and processing between components in the semiconductor device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIGS. 1A through 1C illustrate stages of forming a split via structure for semiconductor packaging in accordance with embodiments of the present technology.
- (2) FIGS. 2A through 2C illustrate alternative stages of forming another split via structure for semiconductor packaging in accordance with embodiments of the present technology.
- (3) FIGS. 3A and 3B respectively illustrate a top-down view and a cross-sectional view of another split via structure connected with parallel traces for semiconductor packaging in accordance with embodiments of the present technology.
- (4) FIGS. 4A and 4B respectively illustrate a top-down view and a cross-sectional view of another split via structure connected with parallel traces for semiconductor packaging in accordance with embodiments of the present.
- (5) FIGS. 5A and 5B respectively illustrate top-down views of the split via structures shown in FIGS. 3A and 4A with different traces connections for semiconductor packaging in accordance with embodiments of the present technology.
- (6) FIGS. 6A and 6B illustrate top-down views of the split via structure shown in FIG. 3A with various traces connections for semiconductor packaging in accordance with embodiments of the present.
- (7) FIG. 7 illustrates a method of forming a split via structure for semiconductor packaging according to embodiments of the present technology.
- (8) FIG. 8 is a schematic view of a system that includes a semiconductor device configured according to embodiments of the presented technology.
- (9) The drawings illustrate only example embodiments and are therefore not to be considered limiting in scope. The elements and features shown in the drawings are not necessarily to scale, emphasis instead being placed upon clearly illustrating the principles of the example embodiments. Additionally, certain dimensions or placements may be exaggerated to help visually convey such principles. In the drawings, the same reference numerals used in different embodiments designate like or corresponding, but not necessarily identical, elements.

DETAILED DESCRIPTION

- (10) In conventional via design, each via is typically surrounded by a via land so as to accommodate overlap tolerances with metal contacts disposed above or below its circuit board layer. Even though the vias themselves may be relatively small, the via lands that surround the vias can create a substantial footprint and occupy a significant amount of surface area on the circuit board. Most conventional circuit boards now implement design rules that require a minimum via land diameter corresponding to the overlap tolerance, which leads to congested area on the circuit board due to multiple/condensed vias placement.
- (11) Additionally, typical via design limits routing spaces, i.e., individual traces cannot be routed as close together as desired, which results in irregular trace designs. This is particularly challengeable and problematic for electronic components that operate at high-speed data transfer rates because

such electronic components generally use differential pair technology for data transmission over pairs of metal traces that are aligned in parallel and spaced apart from one another. When the parallel traces route through a conventional via in a congested area, they may have to divert or present in irregular traces, e.g., serpentine traces, which may increase their spacing distance and/or fail the trace length matching requirements. This also generates additional challenges to the via design because a slight variation in signal path length could create a discontinuity in the differential impedance. Further, conventional via structure may generate fragments for power signal plane design.

(12) To address these challenges and others, the present technology applies a split via structure for the via design in the semiconductor device packaging. The split via structure includes a pair of split vias, each having a via land in a half-moon shape and a via disposed within the via land. The via lands of the pair of split vias are disposed within a single circular region corresponding to a via land in traditional via design. Specifically, the via lands of the split via structure are disposed in mirror symmetric and are separated by a gap. Each of the via lands has an arcuate edge and a chord edge. The vias disposed within each of the via lands can also be in a half-moon shape, having a radius of curvature smaller than the corresponding via land. Alternatively, the vias can have a circular cross-sectional shape and disposed within corresponding via lands.

(13) In the present technique, the traces connected to the split via structure can have various profiles. For example, a differential pair of traces can be aligned in parallel and be connected to the pair of via lands of the split via structure respectively for differential signaling. The differential pair of traces may have inner edges aligned to the chord edges of corresponding via lands. In another example, a pair of traces can be aligned in a horizontal plane and be connected to the pair of via lands of the split via structure respectively for power signal transmission. Here, the edges of the pair of traces for power signaling may be perpendicular to the chord edge of corresponding via land. Overall, the split via structure described in the present technique can reduce the via land footprints and improve differential pair traces in satisfying the trace length matching requirements. Particularly, the described split via structure can be implemented to circuit board design in enabling routing and placement of vias in tight areas. The fabrication of the split via structure is also compatible with existing substrate processes.

(14) FIGS. 1A through 1C illustrate stages of forming a split via structure **100** for semiconductor packaging in accordance with embodiments of the present technology. The split via structure **100** may be processed by a via hole drilling first process on the substrate **100**. FIG. 1A illustrates a top-down view of the split via structure **100** after forming a pair of via holes **101** and **102** on a substrate **110**. As shown, the via holes **101** and **102** each has a half-moon shape with a first radius of curvature ranging from 25 μm to 50 μm . Specifically, each of the via holes **101** and **102** includes a first arcuate edge and a first chord edge. In this example, the via holes **101** and **102** are entirely disposed within a first circular region having a radius same to the first radius of curvature. The first arcuate edges of the via holes **101** and **102** are overlapped/aligned with the edge of the first circular region. Moreover, the via holes **101** and **102** are disposed oppositely in the first circular region, having their first chord edges aligned in parallel and separated by a gap.

(15) In some embodiments, the pair of via holes **101** and **102** of the split via structure **100** can be formed through forming a hard mask layer above the substrate **110** and then patterning the hard mask layer to form openings corresponding to the half-moon shape of the pair of via holes **101** and **102**. Various etching processes, including laser assisted etching, wet chemistry etching, dry gas etching, and/or plasma assisted etching processes can be implemented to drill the via holes **101** and **102** through the substrate **110**. Those via holes can later be filled by conductive materials, e.g., in stage shown in FIG. 1C, to form the pair of vias **108**. In some other embodiments, conductive materials may be filled into the holes right after they are formed in the substrate **110**, to form a pair of vias in the substrate **110**.

(16) FIG. 1B illustrate a top-down view of the split via structure **100** after forming a pair of via

land openings **103** and **104** as well as a pair of trace openings **105** and **106** on a frontside surface of the substrate **110**. As shown, the via land openings **103** and **104** each has a half-moon shape with a second radius of curvature ranging from 50 μm to 100 μm . Specifically, each of the via land openings **103** and **104** has a second arcuate edge (shown in solid lines and dotted lines) and a second chord edge. Here, each of the via land openings **103** and **104** are entirely disposed within a second circular region having a radius similar to the second radius of curvature. The second arcuate edges (including the solid line and dotted line) of the via land openings **103** and **104** are overlapped/aligned with the edge of the second circular region. Notably, the via land openings **103** and **104** are disposed oppositely in the second circular region, having their second chord edges aligned in parallel and separated by the gap. As shown in FIG. 1B, the via holes **101** and **102** are disposed within the via land openings **103** and **104**, respectively. In particular, the centers of the first circular region and the second circular region may be overlapped. Further, the first chord edges of the via holes **101** and **102** and the second chord edges of the via land openings **103** and **104** may be also overlapped. In this example, the trace openings **105** and **106** are in parallel and connected to the via land openings **103** and **104**, respectively.

(17) In some embodiments, the via land openings **103** and **104** as well as the trace openings **105** and **106** can be formed through forming another hard mask layer above the substrate **110**, followed by a patterning process to form openings on the hard mask layer corresponding to the via land openings and the trace openings. Once the hard mask openings are formed, various etching processes, e.g., laser assisted etching, wet chemistry etching, dry gas etching, and/or plasma assisted etching processes can be performed to form via land openings and the trace openings on the front side surface of the substrate **110**. In some other embodiments, the via land openings **103** and **104** can be formed separately to the trace openings **105** and **106**. For example, multiple rounds of hard mask patterning and substrate etching processes can be performed in sequence to form via land openings and trace openings with various thicknesses.

(18) In some embodiments, the via land openings **103** and **104** may have a same thickness to the trace openings **105** and **106** on the frontside surface of the substrate **110**. The thickness of the via land openings **103** and **104** and trace openings **105** and **106** may range from 10 μm to 30 μm . In some other embodiments, the via land openings **103** and **104** may have a larger thickness, e.g., ranging from 10 μm to 30 μm , compared to that of the trace openings **105** and **106**, e.g., ranging from 5 μm to 10 μm .

(19) In some embodiments, another pair of via land openings and another pair of trace openings may be formed on the backside surface of the substrate **110**. For example, the substrate **110** can be flipped over and above-described processing steps can be repeated on the backside surface of the substrate **110** to form similar via land openings and trace openings on the backside surface of the substrate **110**. Moreover, the dimension and location of the pair of via land openings on both surfaces of the substrate **110** may be identical, wherein the via land openings on the frontside surface and the backside surface of the substrate **110** can be connected by the pair of via holes **101** and **102** there through.

(20) FIG. 1C illustrate a top-down view of the split via structure **100** after filling conductive materials into the via holes, via land openings, and trace openings described in FIGS. 1A and 1B, to form a first split via on the left-hand side and a second split via on the right-hand side. The first split via includes a via **107**, a via land **108**, and a trace **109** connected to the via land **108**, all disposed on the left-hand side of the split via structure **100**. Similarly, the second split via includes a via **117**, a via land **118**, and a trace **119** connected to the via land **118**, all disposed on the right-hand side of the split via structure **100**. In some embodiments, the pair of vias **107** and **117** can be made by different conductive materials to the pair of via lands **108** and **118** as well as the pair of traces **109** and **119**. For example, a hard mask layer can be formed above the front side surface of the substrate **110** and only expose the via holes **101** and **102** for conductive material filling. Once the pair of vias **107** and **117** are formed, they can be covered by another hard mask layer, exposing

the via land openings **103** and **104** as well as trace openings **105** and **106** for filling by another conductive material. In some other embodiments, the pair of vias **107** and **117**, the via lands **108** and **118**, as well as the traces **109** and **119** can be made of a same conductive material, i.e., concurrently filling conductive material into the via holes **101** and **102**, via land openings **103** and **104**, and trace openings **105** and **106**. The pair of vias **107** and **117** and the pair of via lands **108** and **118** as well as the pair of traces **109** and **119** can be made of conductive materials including at least one of copper, tungsten, molybdenum, nickel, titanium, tantalum, platinum, silver, gold, ruthenium, iridium, rhenium, rhodium, or alloys thereof.

(21) As shown in FIG. 1C, the split via structure **100** includes the pair of vias each has a half-moon shape with the first radius of curvature. In addition, each of the pair of vias **107** and **117** has the first arcuate edge and the first chord edge, entirely disposed within the first circular region. The pair of vias **107** and **117** are disposed oppositely to each other in the first circular region, having their chord edges aligned in parallel and overlapped with the chord edges of the pair of via lands **108** and **118**. The traces **109** and **119** are respectively connected to the pair of vias **107** and **117** through the pair of via lands **108** and **118**.

(22) FIGS. 2A through 2C illustrate stages of forming another split via structure **200** for semiconductor packaging in accordance with embodiments of the present technology. The split via structure **200** can be processed by a via hole drilling after process on the substrate **210** and similar to the split via structure **100**. For example, the split via structure **200** includes a first split via having a via **207**, a via land **208**, and a trace **209** connected to the via land **208**, all disposed on a left-hand side of the split via structure **210**. In addition, the split via structure **200** includes a second split via having a via **217**, a via land **218**, and a trace **219** that connected to the via land **218**, all disposed on a right-hand side of the split via structure **210**. FIG. 2A illustrates a top-down view of the split via structure **200** after forming a pair of via land openings **203** and **204** as well as a pair of trace openings **205** and **206** on the frontside surface of the substrate **210**. Similar to the via land openings **103** and **104**, the via land openings **203** and **204** each has an arcuate edge and a chord edge, in a half-moon shape with a radius of curvature ranging from 50 μm to 100 μm . Here, each of the via land openings **203** and **204** can be entirely disposed within a circular region with a radius similar to the radius of curvature. As shown, the trace openings **205** and **206** are connected to the via land openings **203** and **204** at the dotted line interfaces, respectively.

(23) In some embodiments, the via land openings **203** and **204** as well as the trace openings **205** and **206** can be formed through forming a hard mask layer above the substrate **210** followed by a patterning process to form openings in the hard mask layer corresponding to the via land openings and trace openings. Once the hard mask openings are formed, various etching processes, e.g., laser assisted etching, wet chemistry etching, dry gas etching, and/or plasma assisted etching processes can be performed on the frontside surface of the substrate **210** through the hard mask layer to form the via land openings **203** and **204** and the trace openings **205** and **206**. In some other embodiments, the via land openings **203** and **204** can be formed separately to the trace openings **205** and **206**, using multiple rounds of hard mask layer patterning and substrate etching processes.

(24) In some embodiments, the via land openings **203** and **204** may have a same thickness to the trace openings **205** and **206** on the frontside surface of the substrate **210**. The thickness of the via land openings and trace openings may range from 10 μm to 30 μm . In some other embodiments, the via land openings **203** and **204** may have a larger thickness, e.g., ranging from 10 μm to 30 μm , compared to that of the trace openings **205** and **206**, e.g., ranging from 5 μm to 10 μm .

(25) FIG. 2B illustrate a top-down view of the split via structure **200** having via holes **201** and **202** respectively formed within the via land openings **203** and **204**, and through the substrate **210**. Similar to the via holes **101** and **102**, the via holes **201** and **202** each has a half-moon shape with a radius of curvature ranging from 25 μm to 50 μm . Each of the via holes **201** and **202** has an arcuate edge and a chord edge. As shown, the via holes **201** and **202** are disposed oppositely in the split via structure **200**, having their chord edges aligned in parallel and disposed next to each other.

Specifically, the chord edges of the via holes **201** and **202** are overlapped/aligned with the chord edges of the via land openings **203** and **204**, respectively.

(26) In some embodiments, the via holes **201** and **202** can be formed by forming a hard mask layer above the via land openings and trace openings and on the frontside surface of the substrate **210**. The hard mask layer can be patterned to form openings corresponding to the via holes **201** and **202**. Various etching processes including laser assisted etching, wet chemistry etching, dry gas etching, and/or plasma assisted etching processes can be then implemented to etch the via holes **201** and **202** on the frontside surface of the substrate **210**.

(27) FIG. 2C illustrate a top-down view of the split via structure **200** after forming the vias **207** and **217**, the via lands **208** and **218**, and the traces **209** and **219** by conductive materials filling. In this example, the via holes **201** and **202** can be filled by a conductive material different to that of the via land openings **203** and **204** and traces openings **205** and **206**. This can be done by implementing a hard mask layer above the frontside surface of the substrate **210** and alternatively open the hard mask layer to expose the via holes for via filling or expose the via land openings and trace openings for the via lands and traces filling.

(28) In some embodiments, another pair of via lands and traces may be formed on the backside surface of the substrate **210**. For example, the substrate **210** can be flipped over and above-described processing steps can be repeated on the backside surface of the substrate **210** to form similar via land openings and trace openings thereon. Moreover, the dimension and location of the pair of via land openings on both surfaces of the substrate **210** may be identical. Similar to the split via structure **100**, the via land openings on the frontside surface and the backside surface of the substrate **210** can be connected by the pair of via holes **201** and **202** disposed there between.

(29) Turning to FIGS. 3A and 3B which respectively illustrate a top-down view and a cross-sectional view of another split via structure **300** connected by differential pair of traces on substrate **330** for semiconductor packaging in accordance with embodiments of the present technology. As shown in FIG. 3A, the split via structure **300** includes a first split via **310** disposed on the left and a second split via **320** disposed on the right. The first split via **310** includes a first via **301** and a first via land **303** surrounding the first via **301**. The first via **301** has a half-moon shape with a first radius of curvature ranging from 25 μm to 50 μm . Specifically, the first via **301** includes an arcuate edge and a chord edge, and is entirely disposed within a first circular region having a radius same to the radius of curvature. In addition, the via land **303** has an arc shape including an outer arcuate edge, an inner arcuate edge, and chord edges disposed at both ends. The via land **303** is entirely disposed within a second circular region, having its outer arcuate edge overlapped with the edge of the second circular region and its inner arcuate edge overlapped with the arcuate edge of the via **301**. As shown in FIG. 3A, a first trace **305** is electrically coupled to the first split via **310** by contacting the first via land **303** through its bottom half outer arcuate edge. In this example, the first trace **305** has two edges, an outer edge being aligned to the arc vertex of the via land **303**'s outer arcuate edge and an inner edge being aligned to the chord edges of the via land **303** and the chord edge of the via **301**.

(30) In this example, the second split via **320** has a similar profile to the first split via **310** and is disposed on the right side of the split via structure **300** in a mirror symmetry to the first split via **310**. As shown, the second split via **320** includes a second via **302** and a second via land **304** surrounding the second via **320**. Similar to the via **301**, the via **302** has a half-moon shape with the first radius of curvature. It also includes an arcuate edge and a chord edge, and is entirely disposed within the first circular region in which the first via **301** also disposed. Similar to the via land **303**, the via land **304** has an arc shape with an outer arcuate edge, an inner arcuate edge, and chord edges disposed at both ends. The via land **304** is also entirely disposed within the second circular region, in which the via land **303** is disposed. Similarly, the via land **304** has its outer arcuate edge overlapped with the edge of the second circular region and its inner arcuate edge overlapped with the arcuate edge of the via **302**. As shown in FIG. 3B, a second trace **306** is electrically coupled to

the second split via **320** by contacting the second via land **304** through its bottom half outer arcuate edge. The second trace **306** also has two edges, an outer edge being aligned to the arc vertex of the via land **304**'s outer arcuate edge and an inner edge being aligned to the chord edges of the via land **304** and the chord edge of the via **302**. In this example, the traces **305** and **310** are aligned in parallel and can be configured to transfer differential signals.

(31) FIG. 3B illustrate the cross-sectional view of the split via structure **300** along the A-A' plane shown in FIG. 3A. This cross-sectional view shows that the split via structure **300** may also include via lands **303'** and **304'** disposed on the other surface of the substrate **330**. Specifically, the vias **301** and **302** each passes through the substrate **330** and interconnects the via lands of each of the first and second split vias, respectively. In this cross-sectional view, the via lands **303** and **303'** and via **301** of the first split via **310** are disposed in symmetry to the via lands **304** and **304'** and via **302** of the second split via **320**. The first split via **310** and the second split via **320** are separated by a gap having a width close to 25 μm . In this example, each of the vias **301** and **302** may have a height ranging from 100 μm to 160 μm and a width close to 50 μm . Further, each of the via lands **303** and **303'** may have a thickness ranging from 10 μm to 30 μm and a width close to 20 μm .

(32) The split via structure **300** can be processed through the stages described in FIGS. 1A through 1C or FIGS. 2A through 2C. For example, as illustrate in the FIG. 1A, the vias **301** and **302** can be firstly fabricated by thrilling via holes through the substrate **330**. The patterning of via lands openings and trace openings on the substrate **330** can be followed up. A single step conductive material filling process or multiple steps conductive materials filling process can be further conducted to finalize the split via structure **300** shown in the FIGS. 3A and 3B. In another example and as illustrated in FIGS. 2A through 2C, the via land openings and trace openings can be firstly processed and followed by a via hole openings process. The vias and via lands of the split via structure **300** can be made of conductive materials including at least one of copper, tungsten, molybdenum, nickel, titanium, tantalum, platinum, silver, gold, ruthenium, iridium, rhenium, rhodium, or alloys thereof.

(33) The split via structure **300** illustrated in FIGS. 3A and 3B can be implemented to via design in advanced semiconductor device packaging. In particular, the individual via lands **303** and **304** of the split via structure **300** has a smaller footprint compared to conventional via lands in circular shape, e.g., having a footprint at least half or less to that of conventional via lands. The reduced footprint of the via lands in the split via structure of the present technology helps save more space on the circuit board, avoiding or reducing irregular traces including serpentine traces that generally are needed to match trace length in traditional via design. The reduced footprint of via lands in the split via structure **300** can also save more room for power and ground planes enhancement on the circuit board. Further, an optimized trace matching between the differential pair, e.g., the traces **305** and **306** in this example, can be achieved by eliminating a far via to via placement requirement through implementing the described split via structure.

(34) FIGS. 4A and 4B respectively illustrate a top-down view and a cross-sectional view of another split via structure **400** that includes micro vias and that is connected by differential pair of traces for semiconductor packaging in accordance with embodiments of the present. The top-down view of the split via structure **400** illustrates a first split via **410** disposed on the left and a second split via **420** disposed on the right. The first split via **410** includes a first micro via **401** and a first via land **403** within which the first micro via **401** is disposed. The first micro via **401** passes through the substrate **410** and has a circular cross-sectional shape with a first diameter. The first via land **403** includes an arcuate edge and a chord edge, and has a half-moon shape with a first radius of curvature. Further, the first via land **403** is entirely disposed within a circular region having the first radius. As shown in FIG. 4A, a first trace **405** is electrically coupled to the first split via **410** by contacting the first via land **403** through its bottom half outer arcuate edge. The second split via **420** includes a second micro via **402** and a second via land **404** which have a same profile and are disposed in mirror symmetric to the first micro via **401** and the first via land **403** of the first split

via **410**, respectively. In particular, the second via land **404** has a half-moon shape and is entirely disposed within the circular region in which the first via land **403** is also disposed. Moreover, a second trace **406** is electrically coupled to the second split via **420** by contacting the second via land **404** through its bottom half outer arcuate edge. In this example, the traces **405** and **406** are in parallel, each having an outer edge being aligned to the arc vertex of the via land **403** or **404** and an inner edge being aligned to the chord edges of the via lands **403** or **404**. Specifically, each of the traces **405** and **406** may have a width same to a lateral width of the via lands **403** and **404** which is along an axis perpendicular to their chord edges.

(35) FIG. **4B** illustrate the cross-sectional view of the split via structure **400** along the B-B' plane shown in FIG. **4A**. This cross-sectional view shows that the split via structure **400** may also include via lands **403'** and **404'** disposed on the other surface of the substrate **430**. Specifically, the vias **401** and **402** each passes through the substrate **430** and interconnects the via lands of each of the first and second split vias, respectively. In this cross-sectional view, the via lands **403** and **403'** and via **401** of the first split via **410** are disposed in symmetry to the via lands **404** and **404'** and via **402** of the second split via **420**. The first split via **410** and the second split via **420** are separated by a gap having a width close to 25 μm . In this example, each of the vias **401** and **402** may have a height ranging from 100 μm to 160 μm and a width close to 45 μm . Further, each of the via lands **403** and **403'** may have a thickness ranging from 10 μm to 30 μm and a width close to 70 μm .

(36) In this example, the via split structure **400** may be processed by the via hole drilling first process illustrated in FIGS. **1A** through **1C** or the via hold drilling after process illustrated in FIGS. **2A** through **2C**. Similar to the split via structure **300**, the vias and via lands of the split via structure **400** can be made of a same conductive filling material or different conductive filling materials. The vias and via lands of the split via structure **400** can be made of conductive materials including at least one of copper, tungsten, molybdenum, nickel, titanium, tantalum, platinum, silver, gold, ruthenium, iridium, rhenium, rhodium, or alloys thereof.

(37) Turning to FIG. **5A** which illustrates another top-down view of the split via structure **300** with a different traces connection for semiconductor packaging in accordance with embodiments of the present technology. Specifically, the split via structure **300** is connected by two traces **505** and **506** that are aligned in a same horizontal plane. As described in FIG. **3A**, the split via structure **300** includes a first split via **310** disposed on the left and a second split via **320** disposed on the right. Here, the first trace **505** is electrically coupled to the first split via **310** by contacting the first via land **303** through a first edge location, i.e., a center portion of the via land **303**'s outer arcuate edge. In this example, both edges of the first trace **505** are perpendicular to the chord edge of the via **301**. Similarly, the second trace **506** is electrically coupled to the second split via **320** by contacting the second via land **304** through a second edge location, i.e., a center portion of the via land **304**'s outer arcuate edge. In this example, both edges of the second trace **506** are perpendicular to the chord edge of the via **302**. The first trace **505** and the second trace **506** can be aligned in a same horizontal plane, as shown in FIG. **5A**, and are disposed on the left side and the right side of the split via structure **300**, respectively. Moreover, the first edge location and the second edge location are disposed oppositely in the circular region within which the first and second via lands **303** and **304** are disposed. In some embodiments, the first and second traces **505** and **506** can be configured to transmit power signals. For example, the trace **505** can be used to transfer a negative supply voltage V_{ss} and the trace **506** can be used to transfer a positive supply voltage V_{dd} .

(38) FIG. **5B** illustrates another top-down view of the split via structure **400** with a different traces connection for semiconductor packaging in accordance with embodiments of the present technology. Specifically, the split via structure **400** is connected by two traces **605** and **606** that are aligned in a same horizontal plane. As shown, the split via structure **400** includes a first split via **410** disposed on the left and a second split via **420** disposed on the right. Each of the vias **401** and **402** is in a circular cross-sectional shape and entirely disposed within the via land **403** and **403**, respectively. Here, a first trace **507** is electrically coupled to the first split via **410** by contacting the

first via land **403** through a third edge location, i.e., a center portion of the via land **403**'s outer arcuate edge. In this example, both edges of the first trace **507** are perpendicular to the chord edge of the via land **403**. Similarly, a second trace **508** is electrically coupled to the second split via **420** by contacting the second via land **404** through a fourth edge location, i.e., a center portion of the via land **404**'s outer arcuate edge. In this example, both edges of the second trace **508** are perpendicular to the chord edge of the via land **404**. The first trace **507** and the second trace **508** can be aligned in a same horizontal plane, as shown in FIG. 5B, and disposed on the left side and the right side of the split via structure **400**, respectively. Moreover, the third edge location and the fourth edge location are disposed oppositely in the circular region within which the first and second via lands **403** and **404** are disposed. In some embodiments, the first and second traces **507** and **508** can be configured to transmit power signals. For example, the trace **507** can be used to transfer a negative supply voltage V_{ss} and the trace **508** can be used to transfer a positive supply voltage V_{dd} .

(39) FIGS. 6A and 6B illustrate top-down views of split via structure **300** having various traces connections for semiconductor packaging in accordance with embodiments of the present. FIG. 6A illustrates a pair of traces **603** and **604** that are disposed in parallel and are respectively connected to the first split via **310** and the second split via **320**. In particular, each of the traces **603** and **604** has a width smaller than the lateral width of the via lands **303** and **304**, i.e., the distance from the arc vertex of the via land's outer arcuate edge to the chord edge of embedded via. As shown in FIG. 6A, the trace **603** may be connected to the via land **303** at a first arcuate edge location and laterally aligned to a middle region of the via land **303** along its lateral width. Further, the trace **603** may also include necking regions **601** at both edges of the trace **603** and close to the arcuate edge location, in order to strengthen the interconnection there between. The trace **604** may have a similar profile to the trace **603** and they can be in a mirror symmetric, e.g., in parallel, when connected to the split vias **310** and **320**. For example, the trace **604** may be connected to the via land **304** at a second arcuate edge location and laterally aligned to a middle region of the via land **304** along its lateral width. The first and second arcuate edge locations are opposite in the second circular region in which the via lands **303** and **304** are disposed. The trace **604** also include necking regions **602** that are configured to strengthen the interconnection between the trace **604** and the via land **304**.

(40) FIG. 6B illustrates the split via structure **300** that is connected by two traces **607** and **608** which are aligned in a same horizontal plane. As shown, each of the traces **607** and **608** are perpendicular to the chord line of corresponding via lands. Here, the width of the traces **607** and **608** may be smaller than that of the traces **505** and **506**, e.g., having a width less than the lateral width of the corresponding via lands. Similarly, the traces **607** and **608** may include necking regions **605** and **606** respectively so as to strengthen the interconnection between the trace and corresponding via land. In some other embodiments, the traces **603**, **604**, **607** and **608** described in FIGS. 6A and 6B can be implemented to be connected to other split via structures, e.g., the split via structure **400** which includes a pair of vias in circular cross-sectional shape shown in FIGS. 4A and 4B.

(41) The traces with an adjustable width described in FIGS. 6A and 6B provide a feasibility to increase the trace width or adjust the space between differential pair of traces in order to satisfy various via/trace design rules and applications. For example, the split via structure and applicable traces connections described above in this disclosure can enable NAND flash memory devices, MNAND devices, and/or Low Power Double Data Rate (LPDDR) devices package size scaling, e.g., from current 11.5 mm×13 mm to 9 mm×13 mm. In addition, the smaller via land footprints in the split via structures of the present technology enables a smooth trace design. For example, curved travel paths or serpentine segments of traces can be effectively reduced because of the resulted more estate room on the circuit board.

(42) FIG. 7 illustrates a method **700** of forming a split via structure for semiconductor packaging according to embodiments of the present technology. The method **700** includes providing a first

hard mask layer on a substrate, the first hard mask layer having openings, at **702**. For example, a hard mask layer can be formed above the substrate **210**. The hard mask layer can be patterned to form openings corresponding to the via lands **203** and **204** as shown in FIG. 2A. In some other examples, the hard mask layer can also be patterned to form openings corresponding to the traces **205** and **206**.

(43) The method **700** also includes patterning the substrate through the hard mask layer and plating conductive material in the patterned substrate to form a first via land and a second via land on a surface of a substrate, wherein the first via land has a first footprint in a half moon shape with a first radius of curvature and the second via land has a second footprint in the half moon shape with the first radius of curvature, and wherein the first via land and the second via land are disposed entirely within a circular region having the first radius of curvature, at **704**. For example, various etch processes including laser assisted etching, wet chemistry etching, dry gas etching, and/or plasma assisted etching processes can be performed on the frontside surface of the substrate **210** through the hard mask layer openings to form the via land openings **203** and **204** and the trace openings **205** and **206**. In some other examples, the via land openings **203** and **204** can be formed separately to the trace openings **205** and **206**, using multiple sequences of hard mask layer patterning and substrate etching processes. As shown in FIG. 2A, each of the via lands **203** and **204** has a half-moon shape with a first radius of curvature. Further, both of the via lands **203** and **204** are disposed within a circular region having the first radius. Conductive materials, e.g., copper, can be filled into the via land openings **203** and **204** to form the via lands **208** and **218** shown in FIG. 2C.

(44) In addition, the method **700** includes drilling a first hole and a second hole on the surface of the substrate, the first hole and the second hole each passing through the substrate, at **706**. For example, another hard mask layer can be patterned above the substrate **210** and form hard mask layer openings corresponding to the via holes **201** and **202** as shown in FIG. 2B. The via holes **201** and **202** can be formed by drilling the substrate **210** through the hard mask layer openings. Specifically, the via holes **201** and **202** are disposed within the footprints of via lands **203** and **204**, respectively.

(45) Lastly, the method **700** includes plating conductive material in the first hole and the second hole to form a first via with a second radius of curvature and a second via with the second radius of curvature, respectively, wherein the first via and the second via are respectively disposed within the first footprint and the second footprint, at **708**. For example, conductive material, e.g., copper, can be filled in the via holes **201** and **202** to form vias **207** and **217** shown in FIG. 2C. In particular, the vias **207** and **217** are disposed within the footprints of the via lands **208** and **218**, respectively.

(46) Any one of the semiconductor structures described above with reference to FIGS. 1A-6B can be incorporated into any of a myriad of larger and/or more complex systems, a representative example of which is system **800** shown schematically in FIG. 8. The system **800** can include a semiconductor device **810**, a power source **820**, a driver **830**, a processor **840**, and/or other subsystems or components **850**. The semiconductor device **810** can include features generally similar to those of the semiconductor devices described above, and can therefore include the split via structures and semiconductor device assemblies described in the present technology. The resulting system **800** can perform any of a wide variety of functions, such as memory storage, data processing, and/or other suitable functions. Accordingly, representative systems **800** can include, without limitation, hand-held devices (e.g., mobile phones, tablets, digital readers, and digital audio players), computers, and appliances. Components of the system **800** may be housed in a single unit or distributed over multiple, interconnected units (e.g., through a communications network). The components of the system **800** can also include remote devices and any of a wide variety of computer-readable media.

(47) Specific details of several embodiments of semiconductor devices, and associated systems and methods, are described below. A person skilled in the relevant art will recognize that suitable stages

of the methods described herein can be performed at the wafer level or at the die level. Therefore, depending upon the context in which it is used, the term “substrate” can refer to a wafer-level substrate or to a singulated, die-level substrate. Furthermore, unless the context indicates otherwise, structures disclosed herein can be formed using conventional semiconductor-manufacturing techniques. Materials can be deposited, for example, using chemical vapor deposition, physical vapor deposition, atomic layer deposition, plating, electroless plating, spin coating, and/or other suitable techniques. Similarly, materials can be removed, for example, using plasma etching, wet etching, chemical-mechanical planarization, or other suitable techniques.

(48) In accordance with one aspect of the present disclosure, the semiconductor devices illustrated above could be memory dice, such as dynamic random access memory (DRAM) dice, NOT-AND (NAND) memory dice, NOT-OR (NOR) memory dice, magnetic random access memory (MRAM) dice, phase change memory (PCM) dice, ferroelectric random access memory (Fe RAM) dice, static random access memory (SRAM) dice, or the like. In an embodiment in which multiple dice are provided in a single assembly, the semiconductor devices could be memory dice of a same kind (e.g., both NAND, both DRAM, etc.) or memory dice of different kinds (e.g., one DRAM and one NAND, etc.). In accordance with another aspect of the present disclosure, the semiconductor dice of the assemblies illustrated and described above could be logic dice (e.g., controller dice, processor dice, etc.), or a mix of logic and memory dice (e.g., a memory controller die and a memory die controlled thereby).

(49) The devices discussed herein, including a memory device, may be formed on a semiconductor substrate or die, such as silicon, germanium, silicon-germanium alloy, gallium arsenide, gallium nitride, etc. In some cases, the substrate is a semiconductor wafer. In other cases, the substrate may be a silicon-on-insulator (SOI) substrate, such as silicon-on-glass (SOG) or silicon-on-sapphire (SOP), or epitaxial layers of semiconductor materials on another substrate. The conductivity of the substrate, or sub-regions of the substrate, may be controlled through doping using various chemical species including, but not limited to, phosphorous, boron, or arsenic. Doping may be performed during the initial formation or growth of the substrate, by ion-implantation, or by any other doping means.

(50) The functions described herein may be implemented in hardware, software executed by a processor, firmware, or any combination thereof. Other examples and implementations are within the scope of the disclosure and appended claims. Features implementing functions may also be physically located at various positions, including being distributed such that portions of functions are implemented at different physical locations.

(51) As used herein, including in the claims, “or” as used in a list of items (for example, a list of items prefaced by a phrase such as “at least one of” or “one or more of”) indicates an inclusive list such that, for example, a list of at least one of A, B, or C means A or B or C or AB or AC or BC or ABC (i.e., A and B and C). Also, as used herein, the phrase “based on” shall not be construed as a reference to a closed set of conditions. For example, an exemplary step that is described as “based on condition A” may be based on both a condition A and a condition B without departing from the scope of the present disclosure. In other words, as used herein, the phrase “based on” shall be construed in the same manner as the phrase “based at least in part on.”

(52) As used herein, the terms “top,” “bottom,” “over,” “under,” “above,” and “below” can refer to relative directions or positions of features in the semiconductor devices in view of the orientation shown in the Figures. These terms, however, should be construed broadly to include semiconductor devices having other orientations, such as inverted or inclined orientations where top/bottom, over/under, above/below, up/down, and left/right can be interchanged depending on the orientation.

(53) It should be noted that the methods described above describe possible implementations, and that the operations and the steps may be rearranged or otherwise modified and that other implementations are possible. Furthermore, embodiments from two or more of the methods may be combined.

(54) From the foregoing, it will be appreciated that specific embodiments of the invention have been described herein for purposes of illustration, but that various modifications may be made without deviating from the scope of the invention. Rather, in the foregoing description, numerous specific details are discussed to provide a thorough and enabling description for embodiments of the present technology. One skilled in the relevant art, however, will recognize that the disclosure can be practiced without one or more of the specific details. In other instances, well-known structures or operations often associated with memory systems and devices are not shown, or are not described in detail, to avoid obscuring other aspects of the technology. In general, it should be understood that various other devices, systems, and methods in addition to those specific embodiments disclosed herein may be within the scope of the present technology.

Claims

1. A semiconductor device assembly, comprising: a substrate; a first split via including a first via land that is disposed on a surface of the substrate and that has a first footprint with a half-moon shape with a first radius of curvature, and a first via that passes through the substrate and that has a second radius of curvature, wherein the first via is disposed within the first footprint; and a second split via including a second via land that is disposed on the surface of the substrate and that has a second footprint with the half-moon shape with the first radius of curvature, and a second via that passes through the substrate and that has the second radius of curvature, wherein the second via is disposed within the second footprint, wherein the first via land and the second via land are disposed entirely within a circular region having the first radius of curvature, and wherein the first via land and the second via land are symmetric in the circular region and isolated by a first gap.
2. The semiconductor device assembly of claim 1, wherein the first via land and the second via land each has a first arcuate edge and a first chord edge.
3. The semiconductor device assembly of claim 2, further comprising a pair of traces including a first trace electrically coupled to the first via land with a trace width and a second trace electrically coupled to the second via land with the trace width.
4. The semiconductor device assembly of claim 3, wherein the first trace is coupled to the first via land at a first edge location of the circular region, wherein the second trace is coupled to the second via land at a second edge location of the circular region.
5. The semiconductor device assembly of claim 4, wherein the pair of traces are a differential pair, in which the first trace and the second trace are in parallel and separated by a second gap.
6. The semiconductor device assembly of claim 5, wherein the first gap and the second gap have a same width, and wherein the pair of traces each has an inner edge substantially aligned with the first chord edge of the first via land and the second via land, respectively.
7. The semiconductor device assembly of claim 6, wherein the trace width of each of the pair of traces is equal to a lateral width along an axis perpendicular to the chord edge of each of the first via land and the second via land.
8. The semiconductor device assembly of claim 5, wherein the first gap is smaller than the second gap, and the trace width of the pair of traces is smaller than a lateral width along an axis perpendicular to the chord edge of each of the first via land and the second via land.
9. The semiconductor device assembly of claim 4, wherein the second edge location is opposite to the first edge location in the circular region.
10. The semiconductor device assembly of claim 4, wherein trace edges of the first trace and the second trace are perpendicular to the first chord edges of the first via land and the second via land, respectively.
11. The semiconductor device assembly of claim 2, wherein the first via and the second via each has a second arcuate edge smaller than the first arcuate edge and a second chord edge shorter than the first chord edge.

12. The semiconductor device assembly of claim 11, wherein the second chord edges of the first via and the second via are overlapped with the first chord edges of the first via land and the second via land, respectively.
13. A semiconductor device assembly, comprising: a substrate; a first split via including a first via land that is disposed on a surface of the substrate and that has a first footprint with a half-moon shape with a first radius of curvature, and a first micro via that passes through the substrate and that has a circular cross-sectional shape with a first diameter, wherein the first micro via is disposed within the first footprint; and a second split via including a second via land that is disposed on the surface of the substrate and that has a second footprint with the half-moon shape with the first radius of curvature, and a second micro via that passes through the substrate and that has the circular cross-sectional shape with the first diameter, wherein the second micro via is disposed within the second footprint, wherein the first via land and the second via land are disposed entirely within a circular region having the first radius of curvature.
14. The semiconductor device assembly of claim 13, further comprising a pair of traces including a first trace electrically coupled to the first via land with a trace width at a first edge location of the circular region and a second trace electrically coupled to the second via land with the trace width at a second edge location of the circular region, wherein the first via land and the second via land each has a first arcuate arc edge and a chord edge.
15. The semiconductor device assembly of claim 14, wherein the first via land and the second via land are symmetric in the circular region and isolated by a first gap, and wherein the pair of traces are a differential pair, in which the first trace and the second trace are in parallel and separated by a second gap.
16. The semiconductor device assembly of claim 15, wherein the first gap and the second gap have a same width, wherein the pair of traces each has an inner edge substantially aligned with the first chord edge of the first via land and the second via land, respectively, and wherein the trace width of each of the pair of traces is equal to a lateral width along an axis perpendicular to the chord edge of each of the first via land and the second via land.
17. The semiconductor device assembly of claim 15, wherein the first gap is smaller than the second gap, and the trace width of the pair of traces is smaller than a lateral width along an axis perpendicular to the chord edge of each of the first via land and the second via land.
18. The semiconductor device assembly of claim 14, wherein the second edge location is opposite to the first edge location in the circular region, and wherein trace edges of the first trace and the second trace are perpendicular to the first chord edges of the first via land and the second via land, respectively.
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